Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. The learning rate was set to 0.0003 with the Adam optimizer.

2 Results

Figure 1 shows the validation accuracy during training.

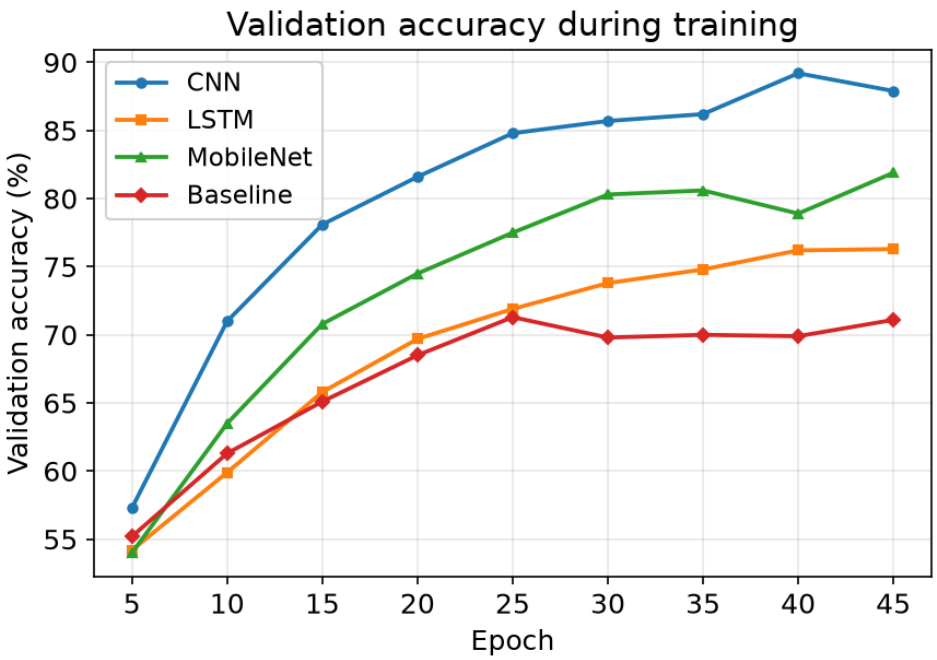


Figure 1: Validation accuracy during training for the models in this experiment.

At epoch 40, MobileNet is about 9 points better than Baseline. LSTM attains a validation accuracy of 56.1% at epoch 5 (Figure 1). As shown in Figure 1, MobileNet attains a validation accuracy of 74.2% at epoch 45.

Figure 2 shows the accuracy by dataset.

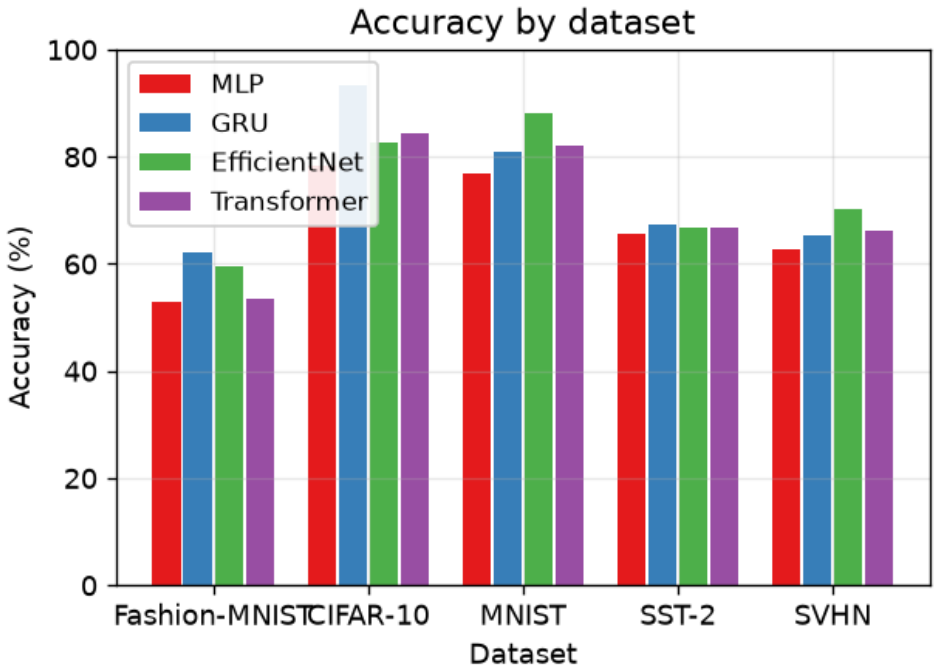


Figure 2: Accuracy by dataset for the models in this experiment.

As shown in Figure 2, GRU achieves 60.0% accuracy on Fashion-MNIST. On MNIST, EfficientNet is 27.2 points better than GRU. On Fashion-MNIST, Transformer obtains a accuracy of 53.0%.

Figure 3 shows the test accuracy during training.

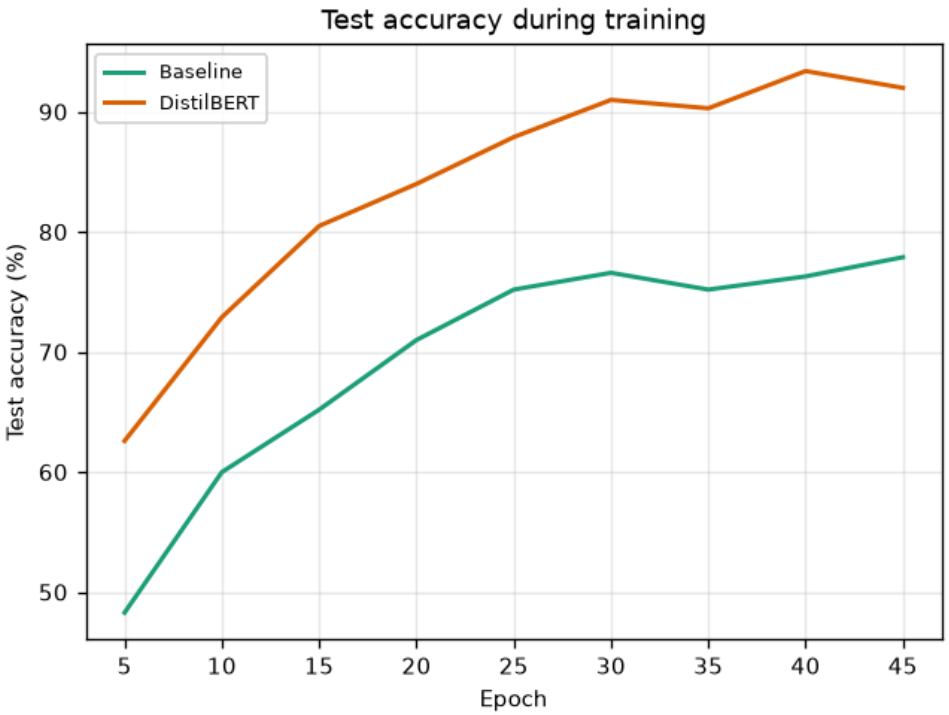


Figure 3: Test accuracy during training for the models in this experiment.

DistilBERT reaches 73% test accuracy at epoch 10. DistilBERT attains a test accuracy of 65.5% at epoch 35. At epoch 15, DistilBERT achieves 75.5% test accuracy.

Figure 4 shows the f1 score by dataset.

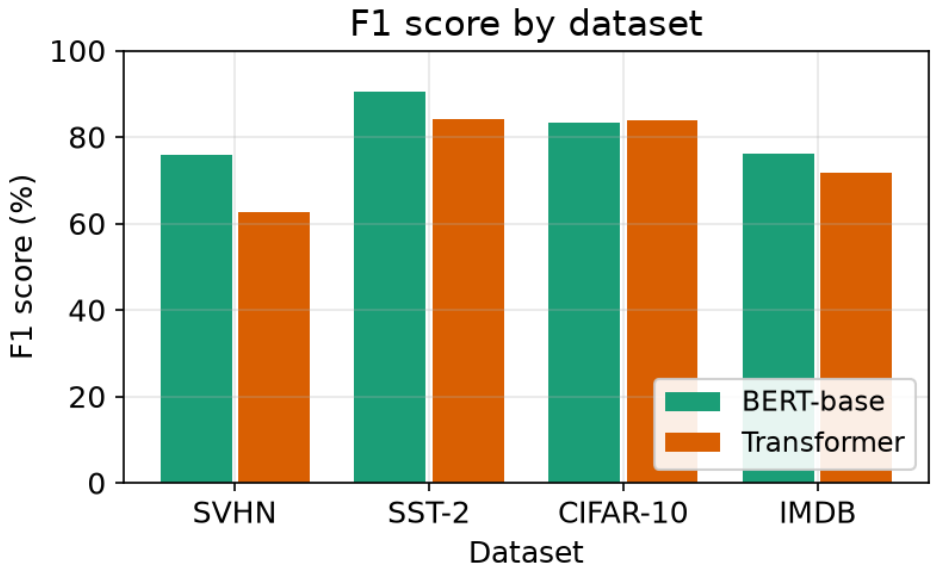


Figure 4: F1 score by dataset for the models in this experiment.

Transformer reaches 71.6% F1 score on IMDB. As shown in Figure 4, on SVHN, BERT-base obtains a F1 score of 76%. BERT-base achieves about 80% F1 score on CIFAR-10.

3 Discussion

On QQP, Random Forest obtains an accuracy of 71.9%. We note that the training was stable for all models.